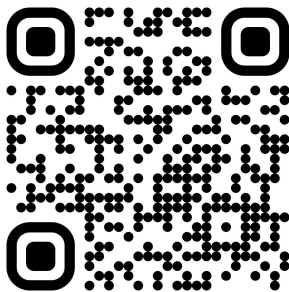


CSSA x DataFest

Beyond The Graph:
Presenting Data with Clarity

CSSA  UC San Diego

Sign In



Scan the QR code to the right or visit:
bit.ly/datafest-beyond-graph



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Welcome!!

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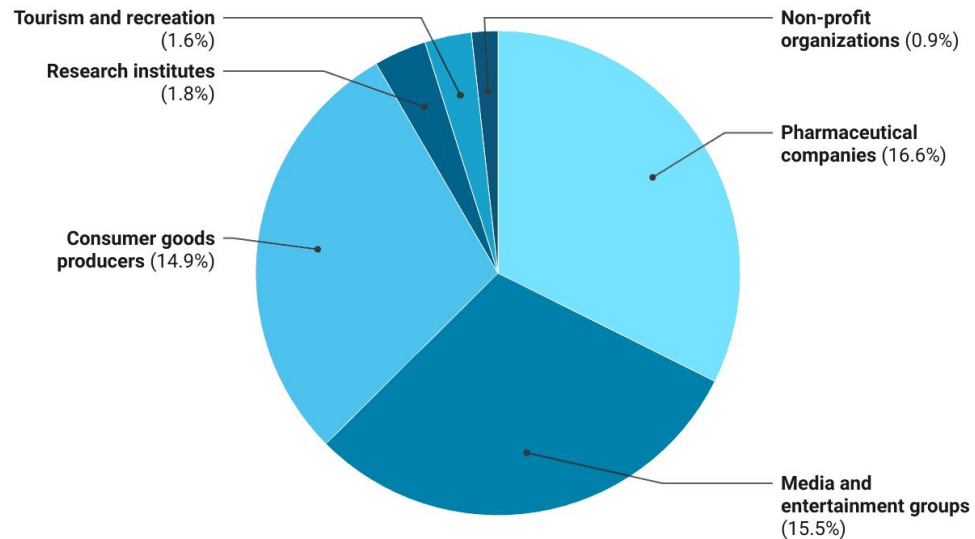
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Put yourself in this scenario:

You have spent hours and weeks on this research project and have written up a very extensive research paper on the details, and have very beautiful graphs that show your findings in this research, but you were later asked to put all that work and information in a 30-50 minute presentation....but your research took you days! How are you supposed to put it in a few simple words and/or graphs and make sure your findings are explained properly?

Market Research Spending According to Industry

Spending in Percentage



(Spending in %)

Source: Market.us Scoop

Why?

→ Context:

- ◆ Assess who your audience is (your boss? hackathon judges?)
- ◆ Do they already know what your project is about?
- ◆ Can you summarize your findings in one sentence or do you need to go into more detail?

Why?

→ Narrative:

◆ Data presentation **is** storytelling.

- Raw data is just a collection of numbers without a plot, presenting gives it a:
 1. Beginning: Setting the scene (**What was our goal?**)
 2. Middle: The conflict (**What challenges/trends did you uncover?**)
 3. End: The resolution (**What action should we take now?**)

Example:

The Fact: Revenue dropped by 10%

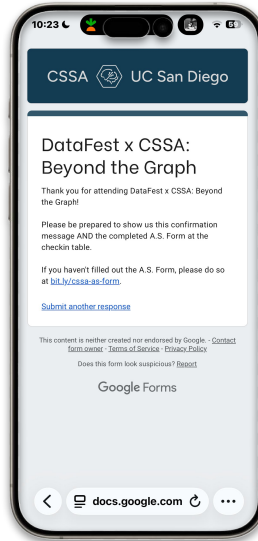
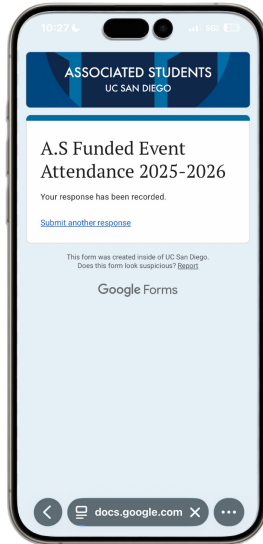
The Story: We are losing the trust of our long-term customers

The audience must feel the impact of the numbers, your job is to make sure it sticks.

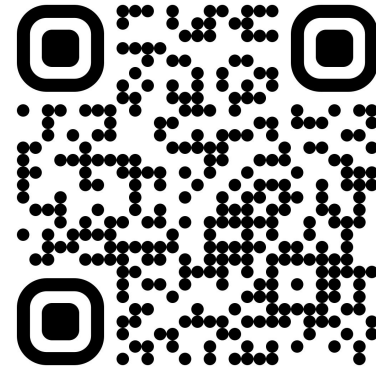
Data Visualization Review

- Comparisons:
 - ◆ Bar/Column Chart, Slope Chart
- Changes / Trends over Time:
 - ◆ Heatmap, Line Graph, Area Chart
- Composition / Part-to-Whole:
 - ◆ Pie Chart, Stacked Bar Chart, Tree Chart
- Distribution:
 - ◆ Box Plot, Histogram
- Relationship & Correlation
 - ◆ Scatter Plot, Bubble Chart
- No Chart
 - ◆ Sometimes a simple number is more powerful than a complex visual.

Make sure to have these two forms available to grab your bev!!



Scan Me!



Gestalt Principles

A law of human perception that describe how our brains group smaller objects together to form larger, meaningful patterns as “wholes” rather than individual parts.

1. **Similarity:** Similar things are grouped together (by color, shape, orientation, etc.)
2. **Proximity:** Things that are near to each other are grouped together
3. **Uniform Connectedness:** Connected regions of visual stimuli are perceived as a single unit
4. **Continuity:** Grouping elements that seem to follow a continuous path in a particular direction
5. **Meaningfulness:** Stimuli form groups if they appear familiar or meaningful
6. **Simplicity:** Each object is reduced to the simplest form possible
7. **Common Region:** Elements in the same defined region tend to be grouped together

Guess the principle



Guess the principle



- **Similarity**
- **Continuity**

Guess the principle



Guess the principle



- **Continuity**
- **Meaningfulness**
- **Uniform**
- **Connectedness(?)**

**Why
does
this
matter?**



Gestalt Principles

Taking advantage of the human perception and altering your presentation with these principles to ensure that the information you need to get across is key to a great presentation. How would you do this, though?

1. **Removing Chart-junk:**

- Remove unnecessary bolding, gridlines, borders, etc. that may distract the audience from your data

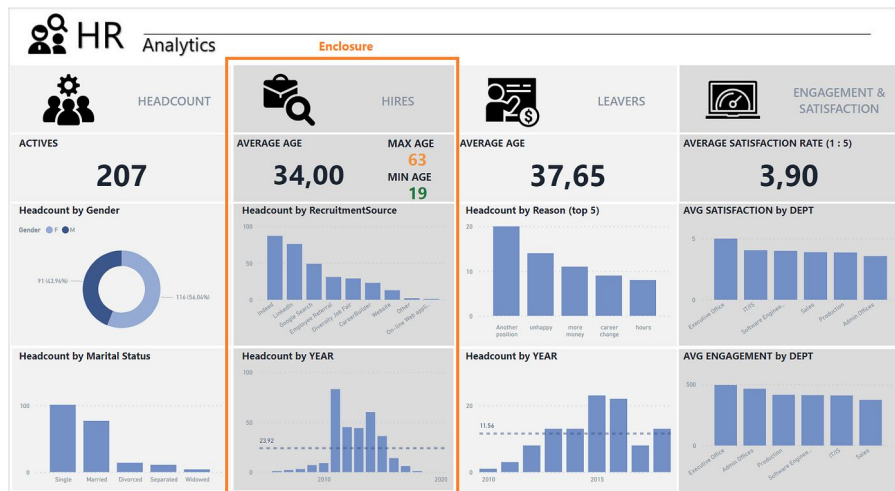
2. **Strategic Use of Colors:**

- Use grey for background data, and a single bold color to highlight specific data points that matter

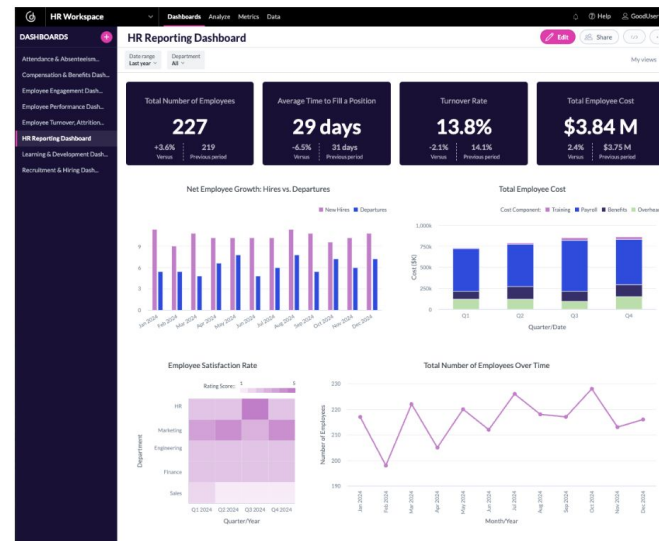
3. **The Squint Test:**

- If you squint at the slide, does the most important part still stand out?

Bad



Good





HR

Analytics

Enclosure



HEADCOUNT

ACTIVES

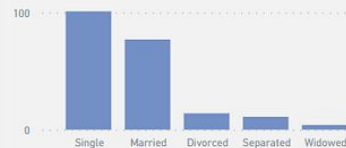
207

Headcount by Gender

Gender ● F ● M



Headcount by Marital Status



HIRES

AVERAGE AGE

34,00

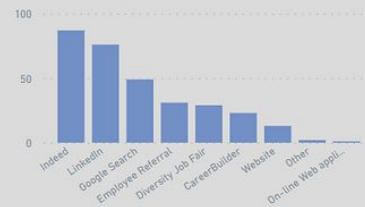
MAX AGE

63

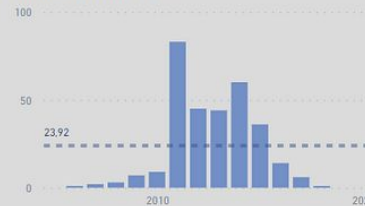
MIN AGE

19

Headcount by RecruitmentSource



Headcount by YEAR

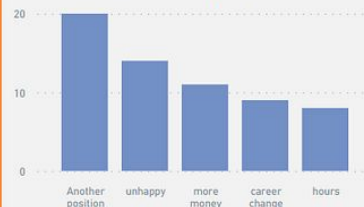


LEAVERS

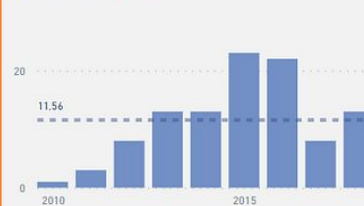
AVERAGE AGE

37,65

Headcount by Reason (top 5)



Headcount by YEAR

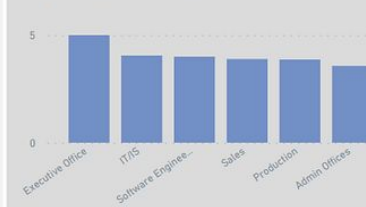


ENGAGEMENT & SATISFACTION

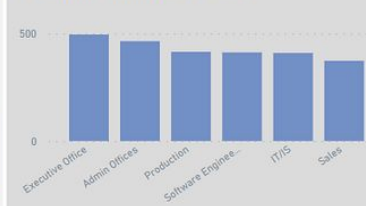
AVERAGE SATISFACTION RATE (1 : 5)

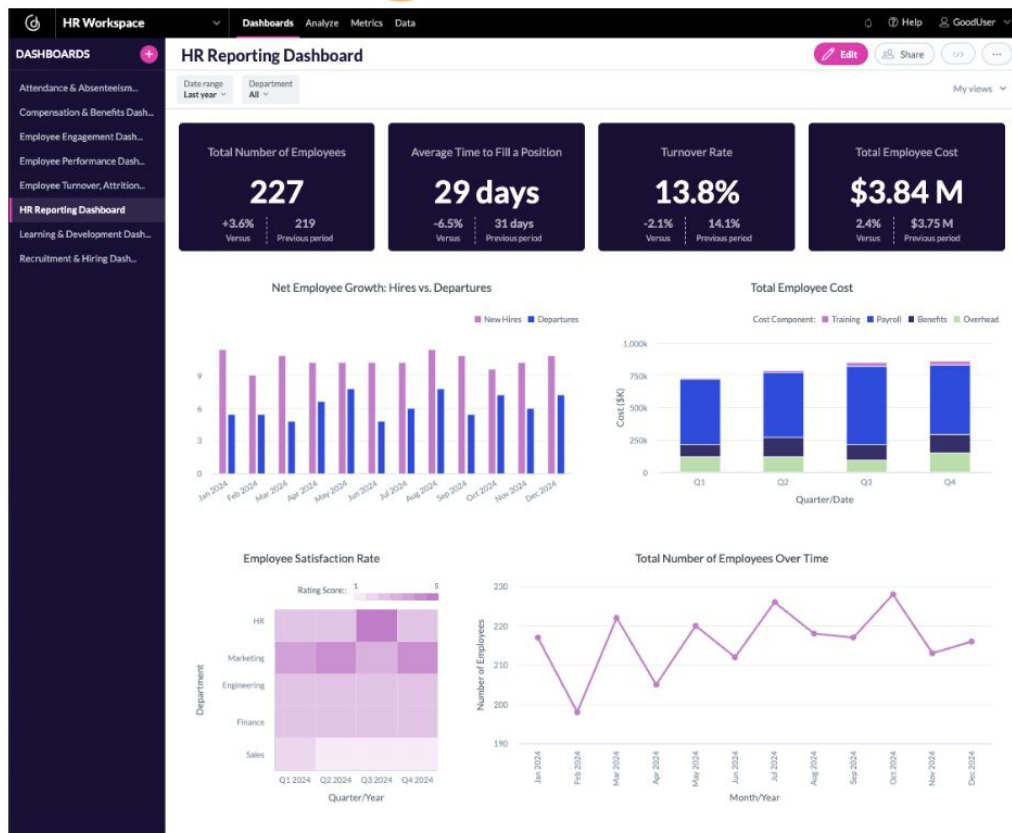
3,90

AVG SATISFACTION by DEPT



AVG ENGAGEMENT by DEPT





**Thank you
for coming!**

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Sign up for DataFest!

